

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY

DEPARTMENT OF ELECTRICAL ENGINEERING

Mini Test February-2025

Course: **B.Tech**

Semester

IV

Branch: **Electrical Engg.**

Subject: **Electrical Machines-II**

Subject Code: **EE-222**

Max Marks: **10**

Question		Marks
Q.No.		
1.	A three phase, 50 Hz, 100 kW induction motor has a full load efficiency of 85%. The stator copper loss and rotor copper loss are each equal to the stator core loss at full load. The mechanical loss is equal to one fourth of the rotor copper loss. Calculate (a) the rotor copper loss (b) the air gap power and (c) the slip.	03
2.	The full load slip of a 500 HP, 50 Hz, three phase induction motor is 0.03. The rotor winding has a resistance of 0.3 ohms/phase. Determine the slip and the power output, if an external resistance of 2 ohms is inserted in each rotor phase. Assume that the torque remains the same.	04
3.	A three phase, 50 Hz, four pole induction motor has a star connected wound motor. The rotor emf is 50 V between the slip rings at standstill. The rotor resistance and standstill reactance are 0.4 ohms and 2 ohms respectively. Calculate: (a) the rotor current per phase at starting with slip rings short circuited. (b) the rotor emf when the motor is running at full load at 1440 rpm. (c) the rotor current at full load (d) rotor power factor at full load.	03